

Final Project Master Repository AUDIT Sindy, Carter, Herve
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2. Proposed Database Schema: Project contains a database scheme with 10 tables managed in SQL:

Table Name	Description
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Patients	Primary demographics: Age, Gender, RegionID.
BloodPressure_Logs	Raw daily readings (Systolic, Diastolic, Pulse).
Biometric_History	Longitudinal tracking of Weight and BMI.
Aggregated_Stats	Pre-computed features: Avg_Systolic, BP_Volatility, Pulse_Pressure.
Clinical_Events	History of Hypertensive_Events, Strokes, or Heart Attacks.
Medication_Records	Active prescriptions to inform Medication Recommendations.
Emergency_Logs	Records of ER visits for training the Emergency Visit Risk model.
Geographic_Regions	Mapping Region codes to socio-economic or environmental data.
Model_Registry	Stores metadata for each of the 15 models (type, version, training date).
Prediction_Results	Final output logs: Risk_Scores, Tiers, and Probabilities.

3. Development Roadmap

Step 1. Our project contains the ten tables listed above and ensures primary and foreign key relationships are strictly defined.

Step 2. A script to generate synthetic data was developed, `data_generator.py` it generates 1200 unique patient entries with varied health information. The information generated adheres to logical trends and contains a `reading_count` history up to 100.

Step 3. A python script containing fifteen models featuring the algorithms logistic regression, random forest classifier, and random forest regressor was developed. It is able to connect to the SQL database and extract necessary features to provide meaningful insights. The models are preprocessed, and split into distinct sets of training/test sets in order to properly run the models.

Step 4. Our developed Streamlit console is able to prove technical viability by providing accuracy metrics such as precision, recall, and RSME. Furthermore, the console is able to pass a dummy patient through the developed system and provide that specific patient's health metrics. The patients results are then saved to the `patients_results_table`.